KN – γ's no polarizados.

$$\frac{d(e\sigma_s)}{d\Omega} = \frac{r_0^2}{2}\left(\frac{\nu'}{\nu_0}\right)^3\left(\frac{\nu_0}{\nu'} + \frac{\nu'}{\nu_0} - s^2\phi\right) \qquad (2.9) \text{ Evans}$$

$$= \frac{r_0^2}{2}\left(\frac{E'_\gamma}{E_\gamma}\right)^3\left(\frac{E_\gamma}{E'_\gamma} + \frac{E'_\gamma}{E_\gamma} - s^2\phi\right)$$

$$\alpha = \frac{E_\gamma}{mc^2} = \frac{h\nu_0}{mc^2}$$

$$= \frac{r_0^2}{2}\left(\frac{1}{1+\alpha(1-c\phi)}\right)^3\left(1+\alpha(1-c\phi) + \frac{1}{1+\alpha(1-c\phi)} - s^2\phi\right)$$

radio clásico de e^-

$$r_0 = \frac{e^2}{4\pi\epsilon_0 mc^2} \sim 2.818 \text{ fm}$$

$$\frac{d\sigma}{d\Omega}\frac{d\Omega}{dE'_\gamma} = \frac{d\sigma}{dE'_\gamma} \qquad \text{Ver } 5.1 \text{ Evans}$$

$$E_\gamma\left[mc^2 - E'_\gamma(1-c\phi)\right] = mc^2 E'_\gamma$$

$$E_\gamma = \frac{mc^2 E'_\gamma}{mc^2 - E'_\gamma(1-c\phi)}$$

$$\left[\frac{d\sigma}{dE'_\gamma}\right] = \left[\frac{d\sigma}{d\Omega}\right]\left[\frac{d\Omega}{dE'_\gamma}\right] = [r_0^2]\frac{1}{[E]} = \frac{m^2}{\text{MeV}} \text{ o } \frac{cm^2}{\text{MeV}} \text{ o } \frac{\text{barn}}{\text{MeV}}$$

KN

Escaneado con CamScanner

$$\frac{d\sigma}{dE'_\gamma} = \underbrace{\frac{r_0^2}{2}\left(\frac{E'_\gamma}{E_\gamma}\right)^3\left(\frac{E_\gamma}{E'_\gamma} + \frac{E'_\gamma}{E_\gamma} - \delta^2\phi\right)}_{\frac{d\sigma}{d\Omega}} \cdot \underbrace{(-2\pi)\frac{mc^2}{E'^2_\gamma}}_{\frac{d\Omega}{dE'_\gamma}}$$

eliminar $\delta\phi$ usando la rel. de dispersión

$$\delta\phi = 1 - \cos\phi = 1 - \left(\frac{mc^2}{E_\gamma} + 1 - \frac{mc^2}{E'_\gamma}\right)^2$$

$$= 1 - \left(\frac{E'_\gamma - E_\gamma}{E_\gamma E'_\gamma} mc^2 + 1\right)^2 = -\left(\frac{E'_\gamma - E_\gamma}{E_\gamma E'_\gamma}\right)^2 m^2c^4 - 2mc^2\frac{E'_\gamma - E_\gamma}{E_\gamma E'_\gamma}$$

$$\frac{E_\gamma + E'_\gamma}{E'_\gamma E_\gamma} + \frac{(E'_\gamma - E_\gamma)^2}{E^2_\gamma E'^2_\gamma} m^2c^4 + 2mc^2\frac{E'_\gamma - E_\gamma}{E_\gamma E'_\gamma}$$ no aparece mucho

$$\frac{E_\gamma}{E'_\gamma} + \frac{E'_\gamma}{E_\gamma} - 1 + \cos^2\phi$$

$$\boxed{\frac{d\sigma}{dE'_\gamma} = (-2\pi)mc^2\frac{r_0^2}{2}\frac{E'_\gamma}{E^3_\gamma}\left(\frac{E_\gamma}{E'_\gamma} + \frac{E'_\gamma}{E_\gamma} + \left[\frac{E'_\gamma - E_\gamma}{E_\gamma E'_\gamma}mc^2 + 1\right]^2 - 1\right)}$$

Klein - Nishina para fotones incidentes no polarizados

KN

Escaneado con CamScanner